

1, Strong Number

A strong number is a number in which the sum of the factorials of its digits equals the number itself.

Definition

A number N is a strong number if:

Sum of factorial of digits of $N = N$.

Examples

Number

calculation

145

$$1! + 4! + 5! = 1 + 24 + 120 = 145$$

Strong?

2.

$$2! = 2$$

yes

405

$$4! + 0! + 5! + 8! + 5! = 40585$$

yes

Not Strong Example

123 →

$$1! + 2! + 3! = 1 + 2 + 6 = 9 \neq 123$$

Python program to check strong number.

```
Num = int(input("Enter a number: "))
```

```
S = 0
```

```
temp = num.
```

```
while temp > 0;
```

```
digit = temp % 10.
```

```
fact = 1
```

```
for i in range(1, digit + 1);
```

```
fact *= i
```

```
S += fact
```

```
temp //= 10.
```



```
if (s == num;
```

```
printf (num, " is a strong number")
```

```
else :
```

```
printf (num, " is not a strong number")
```

2, Perfect Number:

A perfect number is a positive integer that is equal to the sum of its proper divisors, excluding the number itself.

— Example:

6 is a perfect number because its divisors are 1, 2, and 3, and $1 + 2 + 3 = 6$.

Problem Statement:

Given a number, check if it is a perfect number or not.

Sol

```
#include <stdio.h>
```

```
int is perfect (int num) {
```

```
    int sum = 0;
```

```
    for (int i = 1; i < num; i++) {
```

```
        if (num % i == 0) {
```

```
            sum += i;
```

```
        }
```

```
    }
```

```
    return sum == num;
```

```
int main () {
```

```
    int num;
```

```
    printf ("Enter a number: ");
```

```
    scanf ("%d", &num);
```



```
if (is perfect (num)) {  
    printf ("%d is a perfect number\n", num);  
} else {  
    printf ("%d is not a perfect number\n", num);  
}  
return 0;  
}
```

Input : 6

output : 6 is a perfect number.